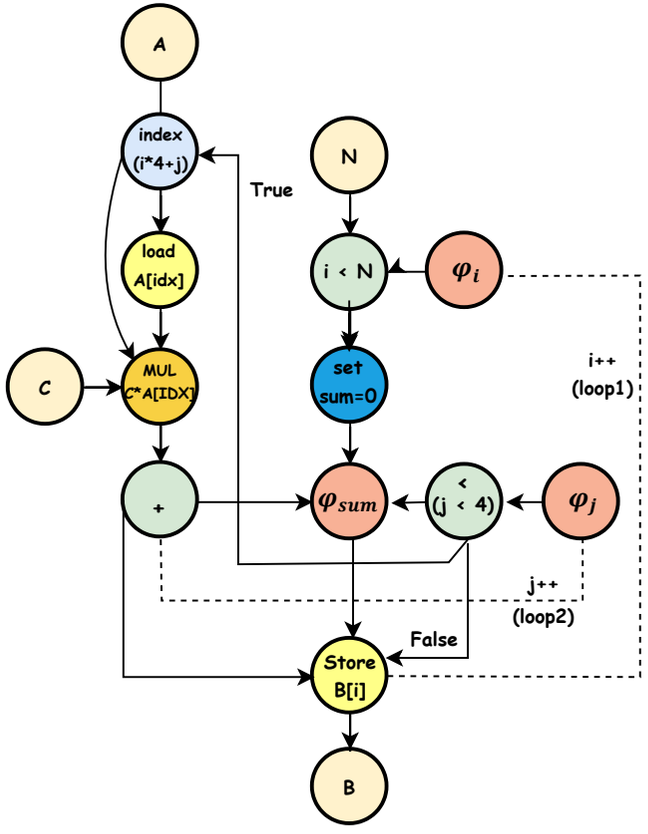
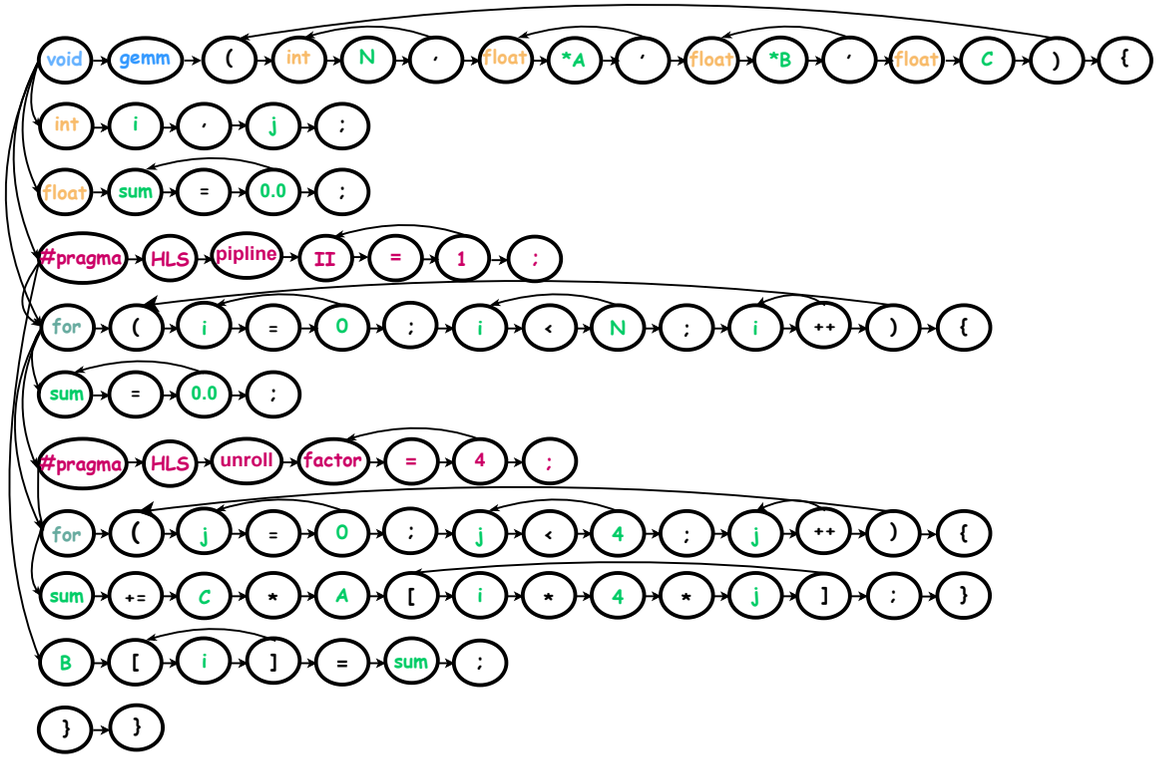
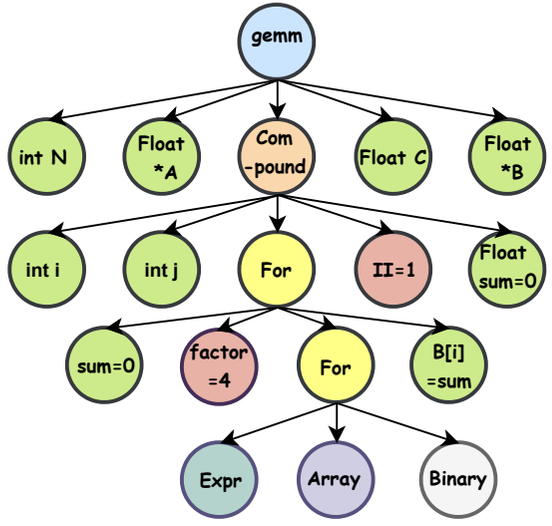

A
index
(i*4+j)
load
A[idx]
C
MUL
C*A[IDX]
+
N
True
i < N
φ_i
set
sum=0
i++
(loop1)
φ_{sum}
<
(j < 4)
φ_j
j++
(loop2)
False
Store
B[i]
B


void gemm (int N , float *A , float *B , float C) {
int i , j ;
float sum = 0.0 ;
#pragma HLS pipline II = 1 ;
for (i = 0 ; i < N ; i ++) {
sum = 0.0 ;
#pragma HLS unroll factor = 4 ;
for (j = 0 ; j < 4 ; j ++) {
sum += C * A [i * 4 * j] ; }
B [i] = sum ;
} }


gemm
int N
Float *A
Com -pound
Float C
Float *B
int i
int j
For
II=1
Float sum=0
sum=0
factor =4
For
B[i] =sum
Expr
Array
Binary